

Demonstration of Proposed Features

Date	23 June 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document summarizes the key features proposed during the planning phase of the **Smart Lender** project and records their implementation and demonstration status. It serves as evidence that the proposed functionalities were successfully developed, integrated, tested, and demonstrated as part of the final project,

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Machine Learning Prediction Pipeline	Developed a complete machine learning pipeline that preprocesses historical loan data, trains four classification algorithms (Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost), evaluates their performance, and automatically selects the best-performing model for deployment.	Implemented	Yes	Successfully demonstrated model training, evaluation, automatic model selection, and generation of serialized model artifacts.
2	Interactive Web Application	Developed a responsive Flask-based web application that allows users to enter applicant information and obtain real-time loan eligibility predictions through an intuitive user interface.	Implemented	Yes	Successfully demonstrated real-time prediction using the loan application form and responsive user interface.
3	AI-Based Prediction Result Analysis	Designed a prediction results page displaying loan approval status, confidence score, AI feature contribution analysis, estimated EMI, Debt-to-Income (DTI) ratio, and complete applicant summary.	Implemented	Yes	Demonstrated explainable AI through feature importance visualization and prediction confidence indicators.
4	EMI and Debt-to-Income (DTI) Calculator	Implemented integrated financial calculators to compute the applicant's monthly EMI and Debt-to-Income ratio based on loan amount, interest rate, tenure, and monthly income.	Implemented	Yes	Successfully demonstrated automatic EMI and DTI calculations for different loan scenarios.
5	Batch Prediction System	Developed a batch processing module that accepts CSV files containing multiple loan applications, predicts loan eligibility for each applicant, and generates a downloadable prediction report.	Implemented	Yes	Demonstrated successful upload, processing, prediction, and download of batch prediction results.
6	Model Performance Dashboard	Developed a dashboard displaying machine learning evaluation metrics, model comparison charts, confusion matrices, ROC curves, feature	Implemented	Yes	Successfully demonstrated visualization of model performance

		importance analysis, and other performance reports generated during training.			reports and comparison among all trained algorithms.
--	--	---	--	--	--

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%